

# Assignment 1: Opinion Network Formation

Three Views of One Survey: Student–Student, Student–Question and  
Question–Question Networks

Team **MATlab**

Mayank Kar   Tanush Garg   Amish Goyal  
IIIT Hyderabad — Dynamical Processes on Complex Networks

September 2026

## 1 Team Name

MATlab

## 2 GitHub Repository

[https://github.com/its-amish/DPCN\\_ass1](https://github.com/its-amish/DPCN_ass1)

The repository contains `clean.py` (shared cleaning), `student_network.ipynb` (Network A), `Extreme Bipartite Network/main.ipynb` (Networks B and C), all figures used below, and the interactive D3 visualisers in `Extreme Bipartite Network/Visualizers/`.

## 3 Dataset Documentation

The survey collects opinions on 60 statements, in four sections of 15: **T** Technology, **E** Education, **S** Society & Ethics, **V** Environment. There are **96 responses**, each answer on a five-point Likert scale, giving at most 5,760 data points.

### 3.1 What the raw answers look like

| Answer            | Count | Share of given answers |
|-------------------|-------|------------------------|
| Strongly Agree    | 2,251 | 43.1%                  |
| Agree             | 1,914 | 36.7%                  |
| Neutral           | 686   | 13.1%                  |
| Disagree          | 256   | 4.9%                   |
| Strongly Disagree | 111   | 2.1%                   |
| Blank             | 505   | —                      |
| “No Comments”     | 37    | —                      |

Two features of this table drive every decision below. First, the class is **overwhelmingly agreeable**: 79.8% of all given answers are Agree or Strongly Agree and only 7.0% are any kind of disagreement. Second, **missing answers are not scattered**, they arrive in whole sections:

- 5 respondents (IDs 44, 60, 68, 73, 78) answered **nothing at all**;
- 4 respondents (IDs 30, 39, 77, 87) answered **only the Technology section**;
- ID 79 skipped all of Environment and ID 123 skipped 10 of its 15 questions;

- every other respondent left at most 4 answers blank.

Blank cells and the 37 “No Comments” answers are treated as missing throughout.

### 3.2 Two preparations of the same data

The three networks need different things from the data, so the project uses two documented preparations.

|                     | Preparation 1 (Network A)                                                                 | Preparation 2 (Networks B, C)                            |
|---------------------|-------------------------------------------------------------------------------------------|----------------------------------------------------------|
| Encoding            | −2 Strongly Disagree ... +2 Strongly Agree; sign = polarity, size = strength              | 1 Strongly Disagree ... 5 Strongly Agree                 |
| Respondents removed | only the 5 completely empty responses → <b>91 students</b>                                | respondents with no extreme answer → <b>89 students</b>  |
| Missing answers     | kept as missing; each pair of students is compared only on questions <i>both</i> answered | dropped from the long-format edge list                   |
| Answers used        | all five levels                                                                           | only Strongly Disagree and Strongly Agree ( $\{1, 5\}$ ) |
| Result              | 91 students × 60 questions, 242 missing cells                                             | 2,362 extreme responses                                  |

Preparation 1 is implemented once in `clean.py` and adds 3 to recover the conventional 1–5 scale, so the two encodings are interchangeable. One caveat should be recorded honestly: the 7 respondents that Preparation 2 drops for having “no extreme view” are mostly not opinion-moderates. Five of them submitted an empty form and one (ID 77) answered only Technology; **ID 119 is the only respondent who genuinely answered the survey without ever using an extreme.** Preparation 2’s remaining 89 students therefore still include three respondents (IDs 30, 39, 87) who answered only 15 of the 60 questions, and their node degrees in Networks B and C are correspondingly understated.

## 4 Pipeline Followed

We build three networks from the same survey, each answering a different question.

|            | Network A                | Network B                  | Network C                  |
|------------|--------------------------|----------------------------|----------------------------|
| Nodes      | 91 students              | 89 students + 60 questions | 60 questions               |
| Edge means | “agree more than chance” | “holds an extreme view on” | “co-held extreme views”    |
| Type       | weighted, one-mode       | bipartite, unweighted      | weighted projection of B   |
| Edges      | 2,026                    | 2,362                      | 1,770 (complete $K_{60}$ ) |
| Author     | Mayank Kar               | Tanush Garg                | Amish Goyal                |

### 4.1 Network A — student–student agreement

**Step 1: a simple agreement score.** For every question both students answered, the pair scores 2 for an identical answer (including both Neutral), 1 for different answers on the same side of the scale (Agree/Strongly Agree, or Disagree/Strongly Disagree), and 0 otherwise. Averaging over the  $n_{ij}$  questions both answered and halving gives a score in  $[0, 1]$ :

$$w_{ij}^{\text{simple}} = \frac{1}{2n_{ij}} \sum_{q \in \text{shared}} s_q(i, j).$$

Averaging rather than summing matters because the Technology-only students share just 15 questions with everyone else.

**Step 2: adjusting for chance.** The simple score never reaches 0 (its minimum over 4,095 pairs is 0.133), so a network built from it links *every* student to every other. The cause is the answer distribution above: when four in five answers are on the agree side, two students drawn at random already match most of the time. We therefore subtract, for each question, the average score  $E_q$  over *all* pairs of classmates — what two randomly chosen students would score on that question:

$$w_{ij} = \frac{1}{n_{ij}} \sum_{q \in \text{shared}} (s_q(i, j) - E_q).$$

A positive weight now means “this pair agrees more than two random classmates would”. Matching on a question the whole class agrees on adds almost nothing; matching on a question that splits the class counts heavily. **Network A links every pair with  $w_{ij} > 0$ :** 2,026 edges, 49.5% of all pairs, one student isolated.

**Step 3: sparsification.** At 49.5% density the network is unreadable and its path lengths are uninformative, so we thin it two ways: a global cutoff swept from weak to strong (experiment 1), and keeping each student’s **two strongest edges** (experiment 2), which yields a 155-edge network used for drawing and for the sparse half of the measurements.

## 4.2 Network B — student–question extremes (bipartite)

Responses are melted into (student, question, response) triples and filtered to  $r \in \{1, 5\}$ , so an edge exists exactly when a student holds an *extreme* view on a question. Moderate and neutral answers are deliberately discarded: the aim is to map ideological intensity rather than consensus. The result is a verified bipartite graph  $G = (U, V, E)$  with  $|U| = 89$ ,  $|V| = 60$ ,  $|E| = 2,362$ , bipartite density 0.4423, and a single connected component.

## 4.3 Network C — question–question projection

Network B is projected onto the question set:

$$e_{q_i, q_j} \in E_Q \iff \exists u \in U : (u, q_i) \in E \wedge (u, q_j) \in E,$$

i.e. two questions are linked when at least one student holds extreme views on both. The unweighted projection turns out to be the complete graph  $K_{60}$ , so all further analysis uses the **weighted** projection  $G_Q^W$ , where  $w_{ij}$  counts the students who hold extreme views on both questions.

## 5 Analysis and Visualizations

### 5.1 Network A: student–student agreement

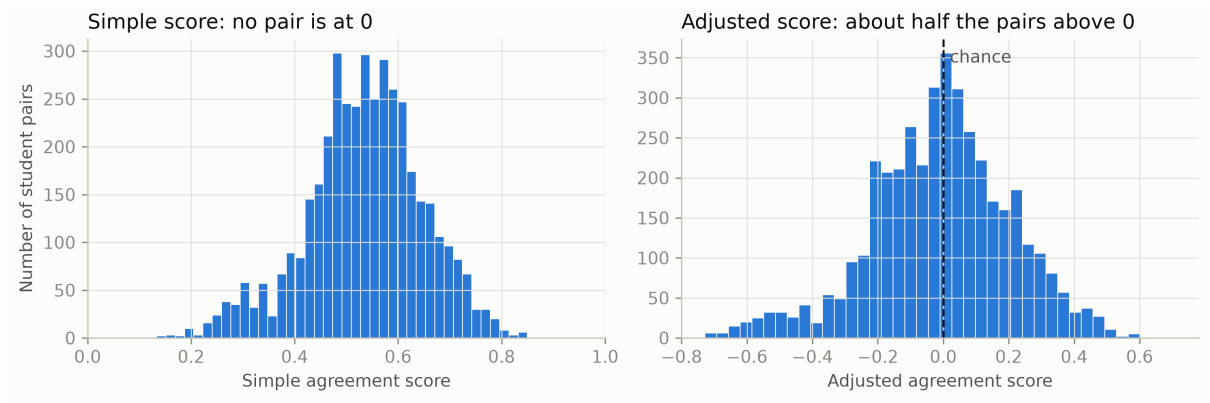


Figure 1: Left: the simple agreement score over all 4,095 pairs — no pair sits at 0, so this score cannot define a network. Right: after subtracting chance agreement the scores centre on 0, and “above chance” becomes a meaningful cutoff. 49.5% of pairs survive it.

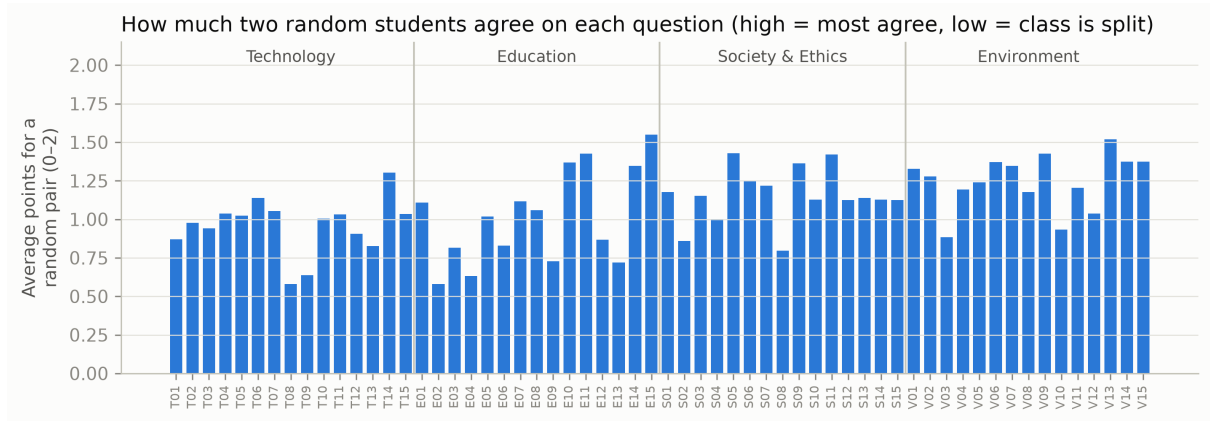


Figure 2: Chance agreement  $E_q$  per question. High bars are consensus questions, where matching answers say little about a pair; low bars are the questions that actually separate students. The most divisive are E02 (written exams measure knowledge,  $E_q = 0.58$ ), T08 (AI diagnosis, 0.58), E04 (online learning, 0.63) and T09 (autonomous vehicles, 0.64); the most consensual are E15 (1.55), V13 (1.52), S05, E11 and V09 (1.43).

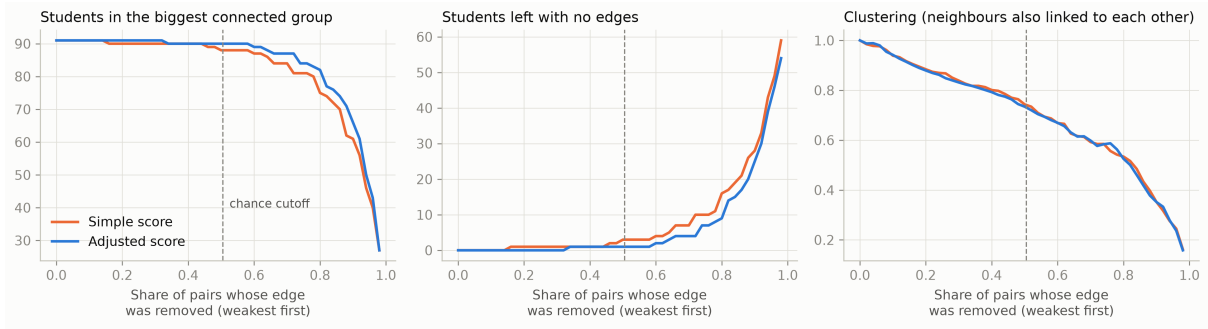


Figure 3: Removing the weakest edges step by step, comparing the simple and adjusted scores at equal density. The biggest connected group erodes gradually — there is no percolation-style collapse. The first students cut off are 120 (no above-chance edge at all), 89, 63 and 22, most of whom answer well below the class average of +1.14. Beyond about 90% removed the curves describe an empty network rather than the class.

Each student's two strongest edges

dot size = total agreement, colour = average answer, labels = first students cut off

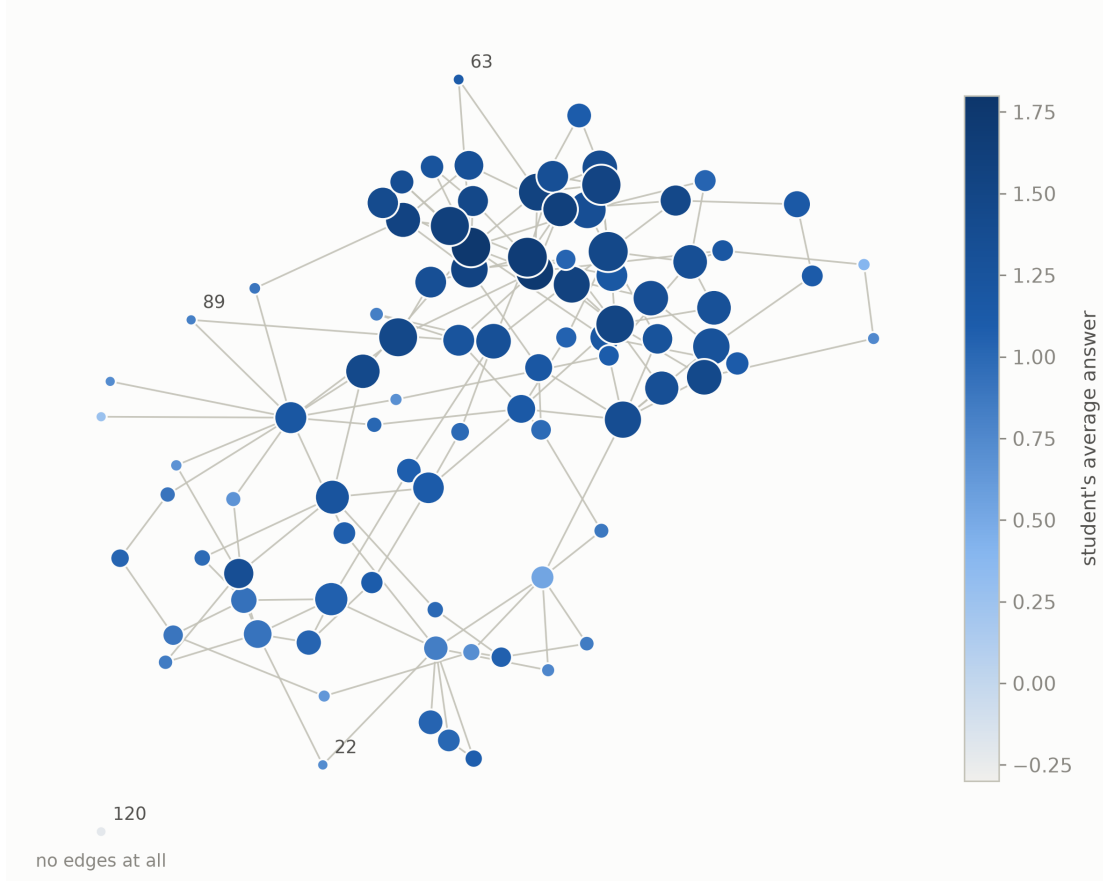


Figure 4: The network reduced to each student's two strongest edges (155 edges, 3.8% of pairs). 90 of the 91 students remain in one piece. Dot size is total agreement (strength) in the full network and colour is the student's average answer: the dark, emphatic answerers form the dense upper core, lighter and more hesitant students sit on the rim, and student 120 has no edge at all.

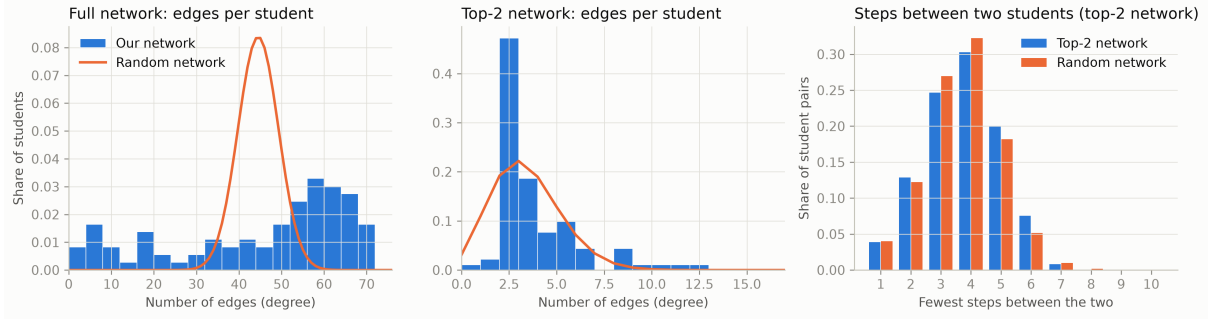


Figure 5: Left and centre: number of edges per student in the full and top-2 networks, against the binomial curve of an ER graph of the same size. Both are far broader than random. Right: distances in the top-2 network are essentially the random values, so the network is short-pathed without being random.

|                              | Full network | ER graph | Top-2 network | ER graph    |
|------------------------------|--------------|----------|---------------|-------------|
| Edges                        | 2,026        | 2,026    | 155           | 155         |
| Density                      | 0.495        | 0.495    | 0.038         | 0.038       |
| Mean degree                  | 44.53        | 44.53    | 3.41          | 3.41        |
| Degree spread ( $\sigma_k$ ) | 21.00        | 4.59     | 2.23          | 1.80        |
| Average clustering           | 0.730        | 0.494    | 0.154         | 0.026       |
| Transitivity                 | 0.757        | 0.494    | 0.090         | —           |
| Average shortest path        | 1.52         | 1.51     | 3.76          | 3.65        |
| Diameter                     | 4            | 2        | 7             | $\approx 5$ |
| Global efficiency            | 0.732        | 0.747    | 0.308         | 0.338       |
| Largest component            | 90           | 91       | 90            | 87.6        |
| Small-world $\sigma$         | <b>1.46</b>  | —        | <b>5.66</b>   | —           |
| Degree assortativity         | 0.027        | -0.023   | -0.323        | -0.023      |

Table 1: Network A against ER graphs with the same number of nodes and edges (each ER column is the mean over 20 such graphs).

**Reading the measurements.** Clustering is far above random in both versions — six times random in the top-2 network, where each student contributed only two edges — while distances match the random values. That is the small-world signature,  $\sigma = 1.46$  in the full network and 5.66 in the sparse one. Degrees are much more spread out than an ER graph allows (21.0 against 4.6), but bounded: with 91 students there is no evidence of a power-law. The negative assortativity of the top-2 network is largely constructed, since students naming the same popular classmate leave that classmate ringed by low-degree neighbours.

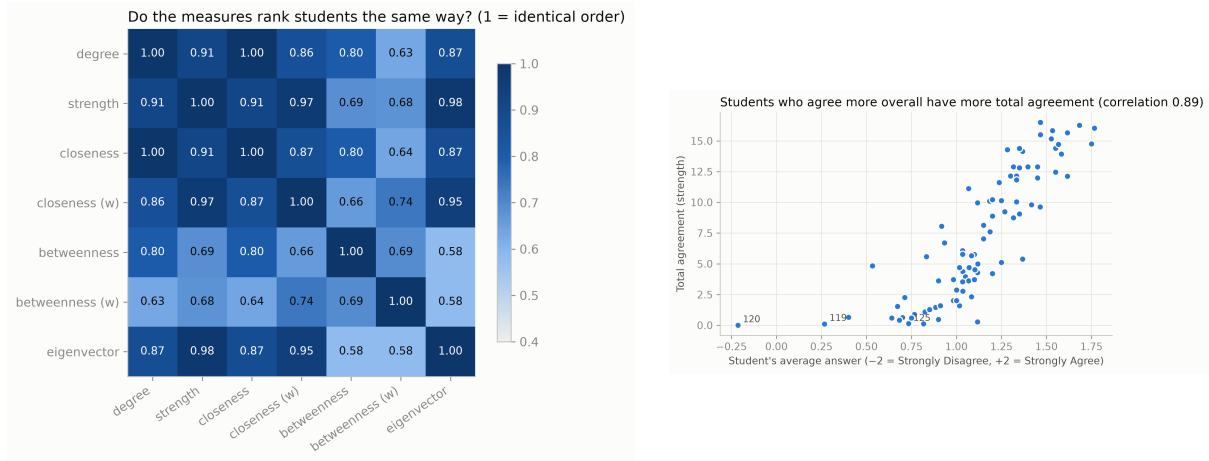


Figure 6: Left: how similarly the centrality measures rank the 91 students. Degree, closeness, strength and eigenvector agree closely ( $\rho \approx 0.85$ – $1.00$ ); betweenness is the odd one out ( $\rho = 0.58$ – $0.80$ ). Right: a student's total agreement tracks their average answer almost perfectly ( $\rho = 0.89$ ).

**Two kinds of central student.** Strength and eigenvector centrality favour the emphatic answers (67, 110, 41, 114, 105), whose edges are the heaviest and mostly with each other. Degree and closeness favour a different set (52, 99, 101, 38, 53) who mix Agree and Strongly Agree and have the *most* edges. The same emphatic core reappears in the top-2 network, so it is not an artefact of density. The periphery is stable across every measure: 120, 125, 119, 89, 22, 63. Betweenness is led by 87 and 77, who answered only 15 questions, so that ranking is unreliable rather than meaningful.

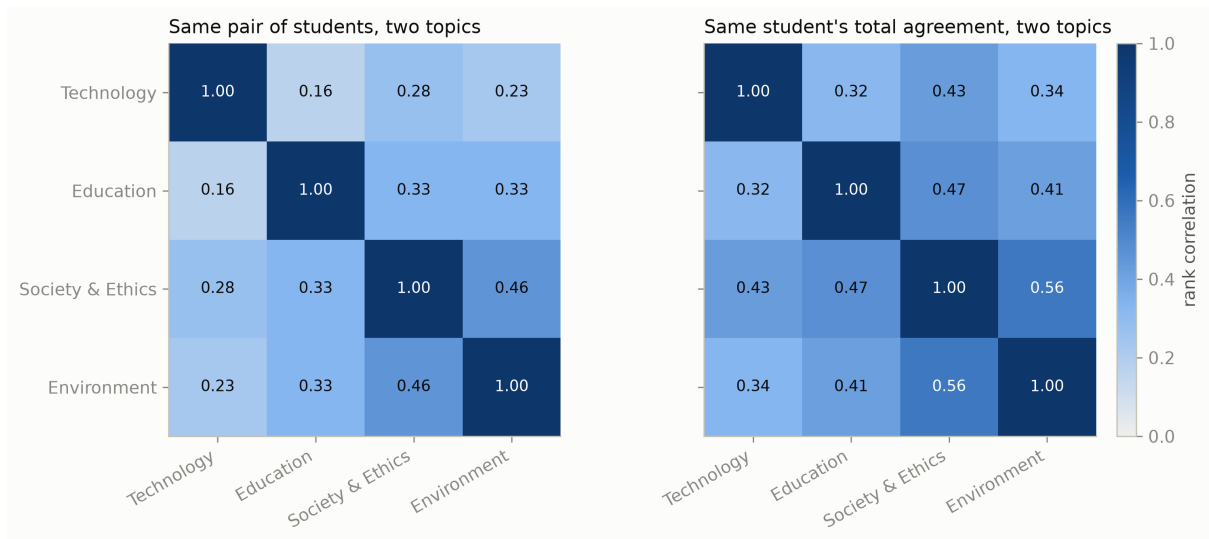


Figure 7: One network per topic. Left: how similarly two topics score the *same pair* of students. Right: how similarly they rank the same student's total agreement. Society and Environment line students up most alike (0.46 and 0.56); Technology and Education least (0.16). Nothing approaches 1.

Each topic alone produces a network much like the whole survey (roughly half of pairs linked, clustering 0.67–0.75), but the correlations above are all well below 1: **who a student agrees with depends on the topic**, and agreement is not one single attitude.

## 5.2 Network B: student–question extremes

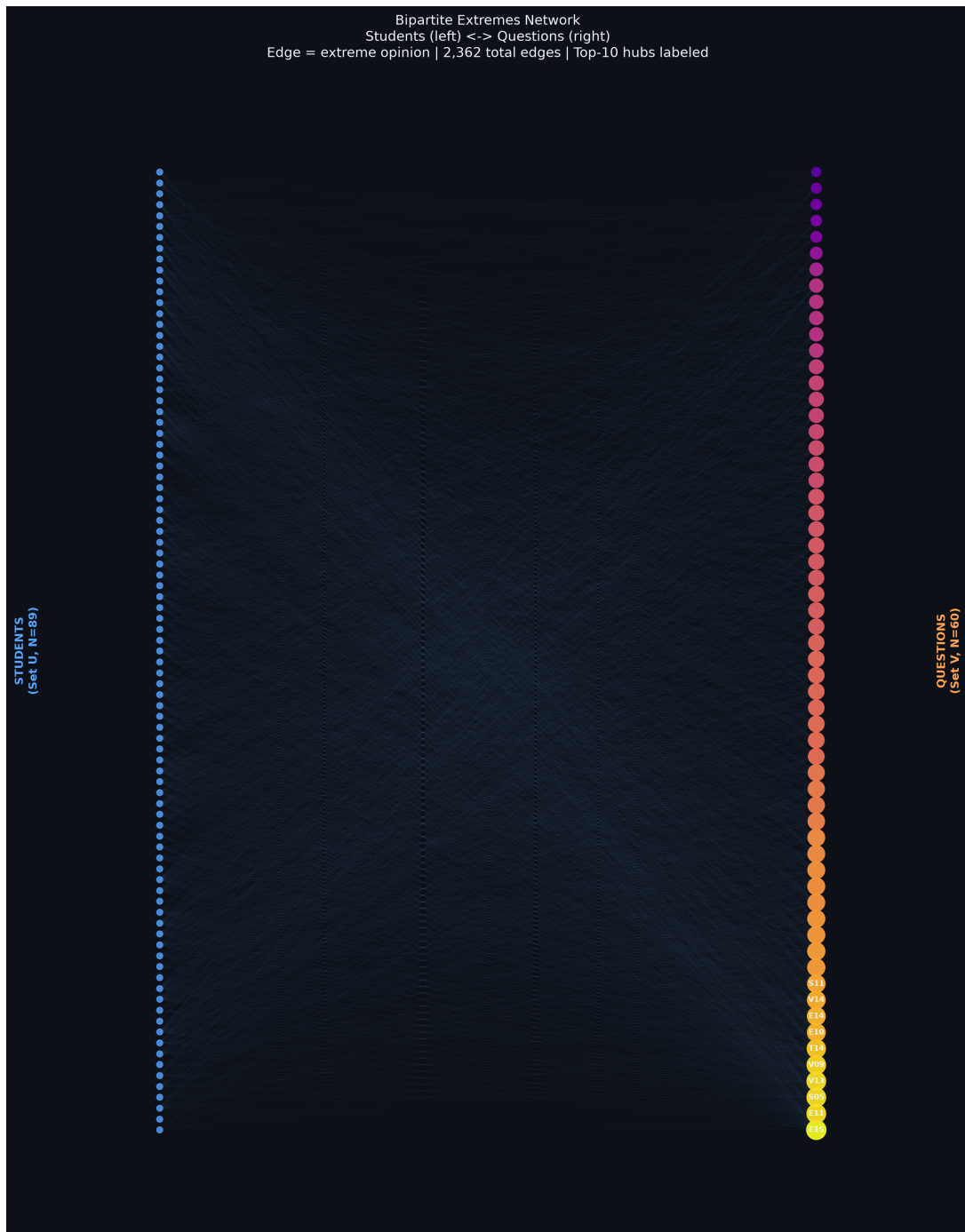


Figure 8: The bipartite extremes network: students (left) joined to the questions (right) on which they hold a Strongly Agree or Strongly Disagree view. Question nodes are coloured by degree; the ten largest “gravity wells” are labelled.

With 2,362 edges over a possible  $89 \times 60 = 5,340$ , the bipartite density is 0.4423: the cohort holds extreme views broadly rather than on a handful of pet topics. The graph has exactly one connected component.

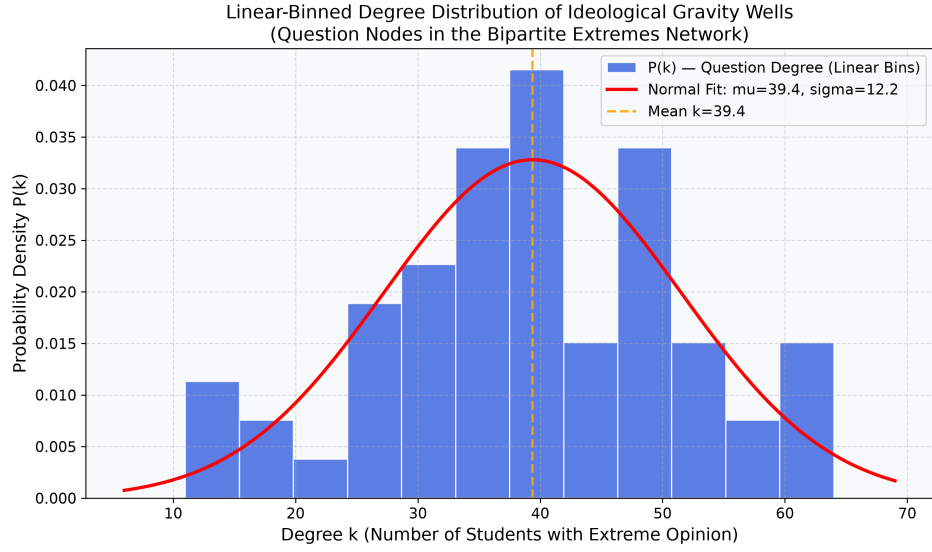


Figure 9: Linear-binned degree distribution of the 60 question nodes with a fitted normal curve ( $\mu = 39.37$ ,  $\sigma = 12.16$ ; range 11–64). The peaked, bounded shape rules out the monotonic decay of a power law.

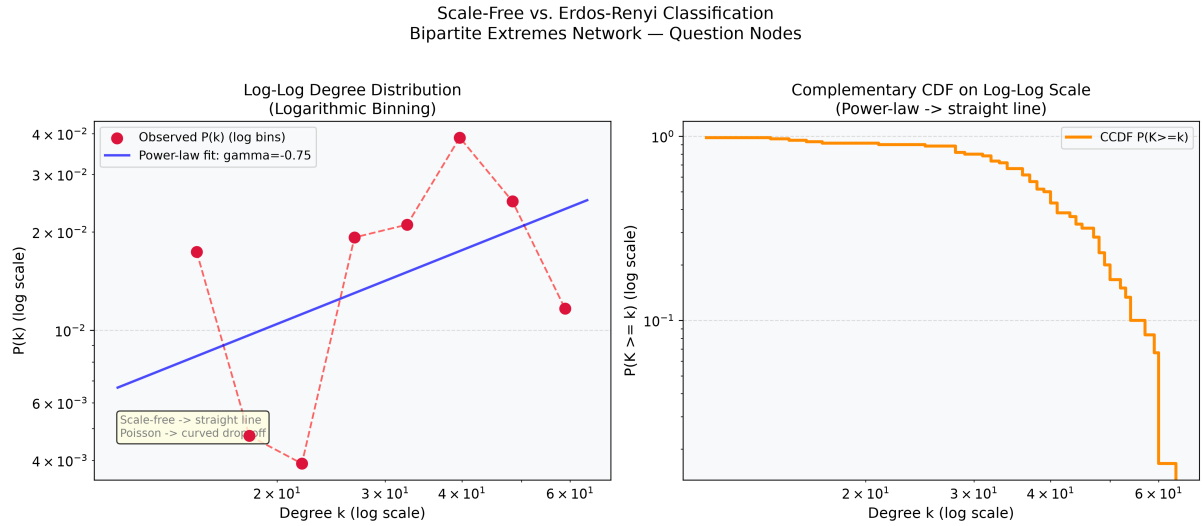


Figure 10: Log-log degree distribution with logarithmic binning (left) and the complementary CDF (right). The best-fit exponent  $\hat{\gamma} = 0.750$  lies far outside the  $2 < \gamma < 3$  band required of a scale-free network, and both curves bend rather than run straight.

The distribution is not Poisson either: the dispersion index  $\text{Var}(k)/\langle k \rangle = 147.9/39.4 \approx 3.75$  is nearly four times the ER expectation of 1. The extremes network is **bounded but overdispersed** — no mega-hubs, yet some questions attract far more intensity than chance allows.

### 5.3 Network C: question-question projection

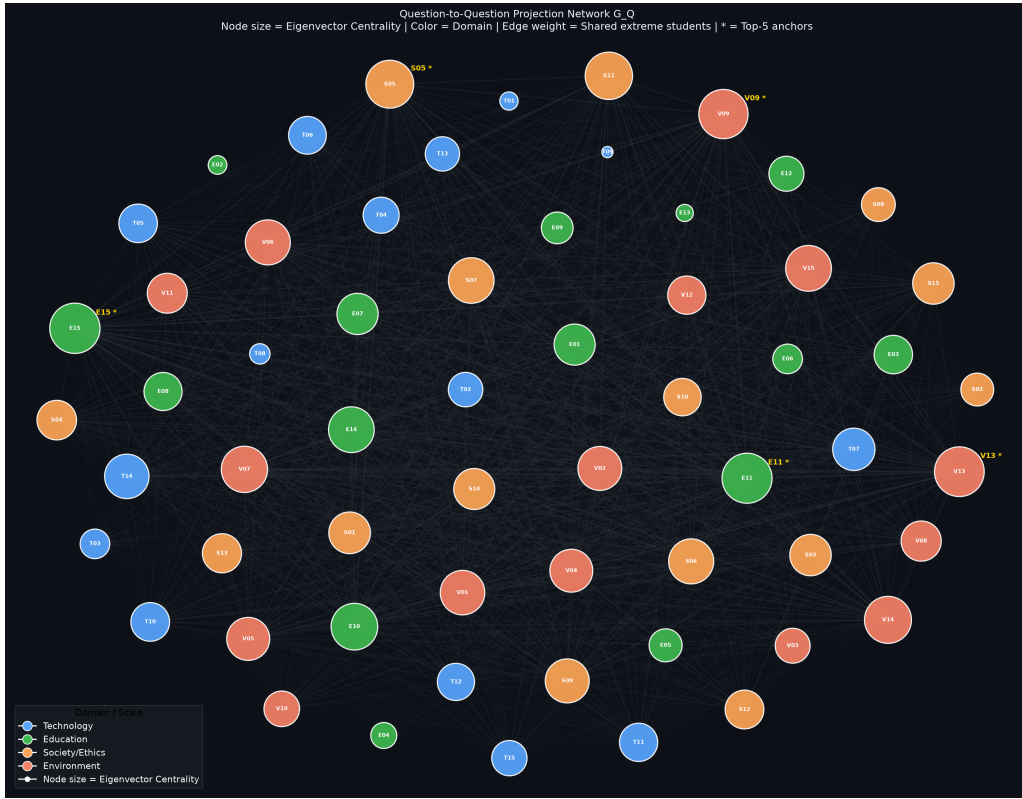


Figure 11: The question-question projection. Every pair of questions is joined, i.e. the unweighted projection is the complete graph  $K_{60}$  (1,770 edges, density 1.0, clustering 1.0, average path length 1.0). Colour marks the four survey domains.

Completeness is itself the finding: **for every pair of questions there is at least one student holding extreme views on both**, so extreme opinion in this cohort is not organised into subject-specific silos. It also means unweighted centrality is uniform, which is why the weighted projection  $G_Q^W$  carries the analysis.

Centrality Analysis: Identifying Ideological Gravity Wells  
(Top 15 Questions in the Weighted Projected Network  $G_Q^W$ )

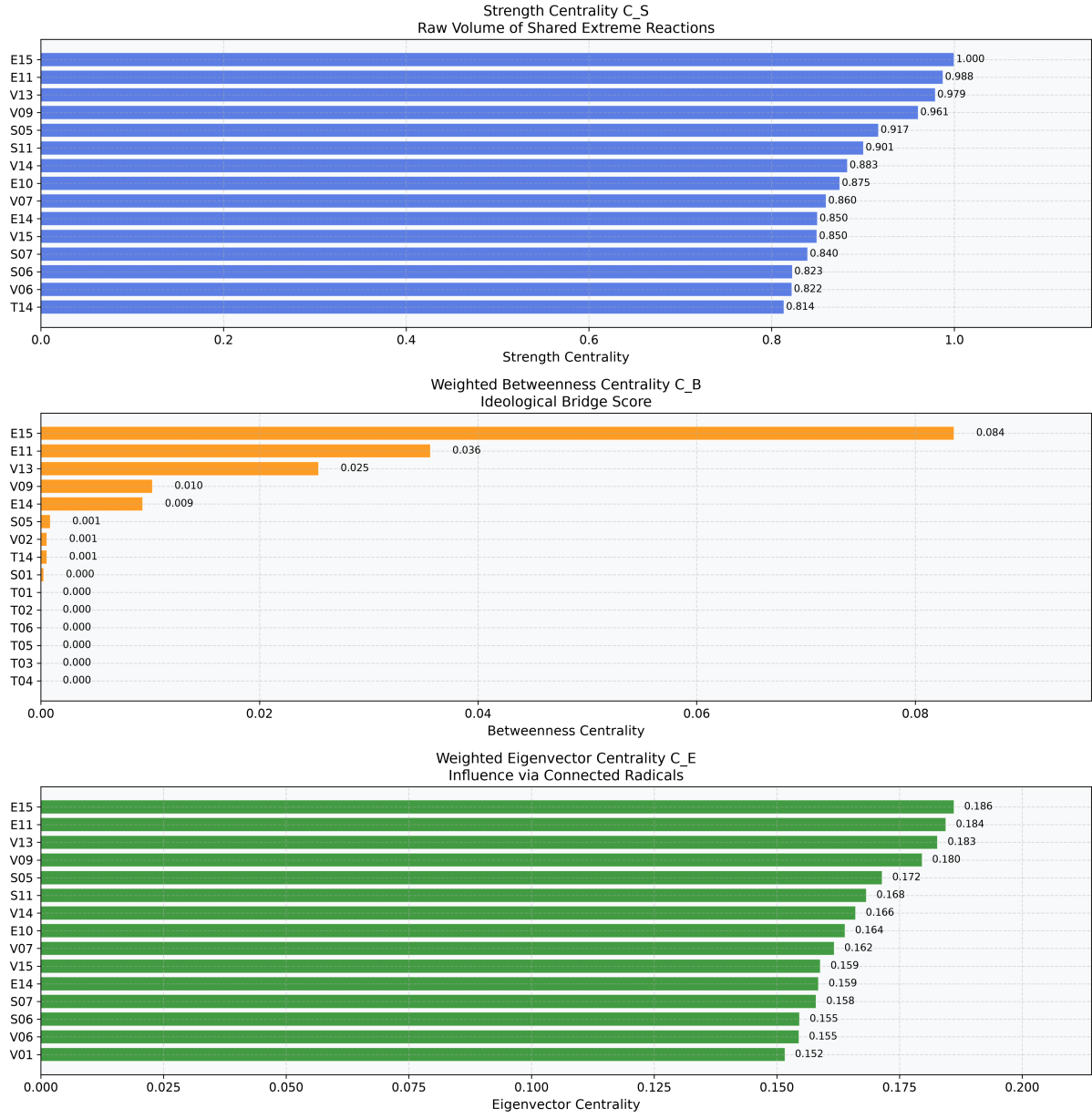


Figure 12: Strength, weighted betweenness and weighted eigenvector centrality on  $G_Q^W$ . Strength and eigenvector decay gradually; betweenness is sharply skewed, collapsing after the top five questions.

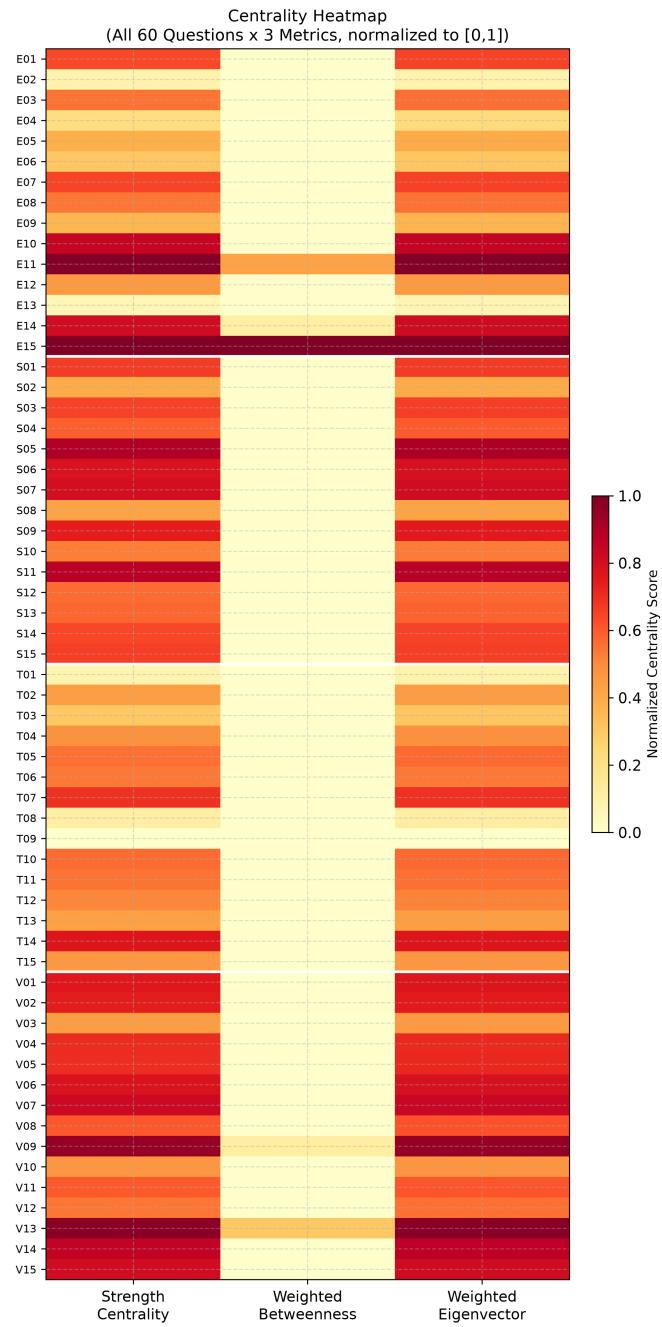


Figure 13: All 60 questions across the three weighted measures. The hot rows are the questions that act as ideological anchors.

| ID  | Statement                                                                        | Strength | Eigenvector |
|-----|----------------------------------------------------------------------------------|----------|-------------|
| E15 | Continuous learning and skill development are essential throughout one's career  | 1.000    | 0.186       |
| E11 | University curricula should be updated more frequently for emerging technologies | 0.988    | 0.184       |
| V13 | Companies should be accountable for their environmental impacts                  | 0.979    | 0.183       |
| V09 | Products should be designed for reuse, repair and recycling                      | 0.961    | 0.180       |
| S05 | Individuals should control how their personal data are collected and used        | 0.917    | 0.172       |
| S11 | People should be free to express differing opinions (absent harm)                | 0.901    | 0.168       |
| V14 | International cooperation is necessary for environmental challenges              | 0.883    | 0.166       |
| E10 | Universities should prioritise innovation over rote learning                     | 0.875    | 0.164       |

Table 2: The strongest ideological anchors in  $G_Q^W$ . Weighted betweenness ranks the same three questions first (E15 0.084, E11 0.036, V13 0.025) before collapsing towards zero.

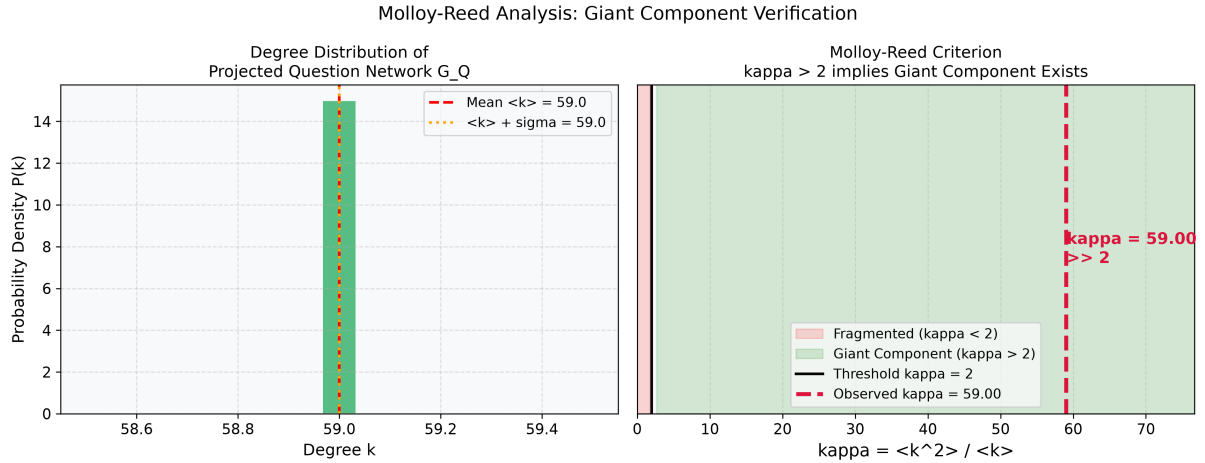


Figure 14: Molloy-Reed criterion on the unweighted projection. Every node has degree 59, giving  $\kappa = \langle k^2 \rangle / \langle k \rangle = 59 \gg 2$ : the giant component is guaranteed, by the widest possible margin for a 60-node graph.

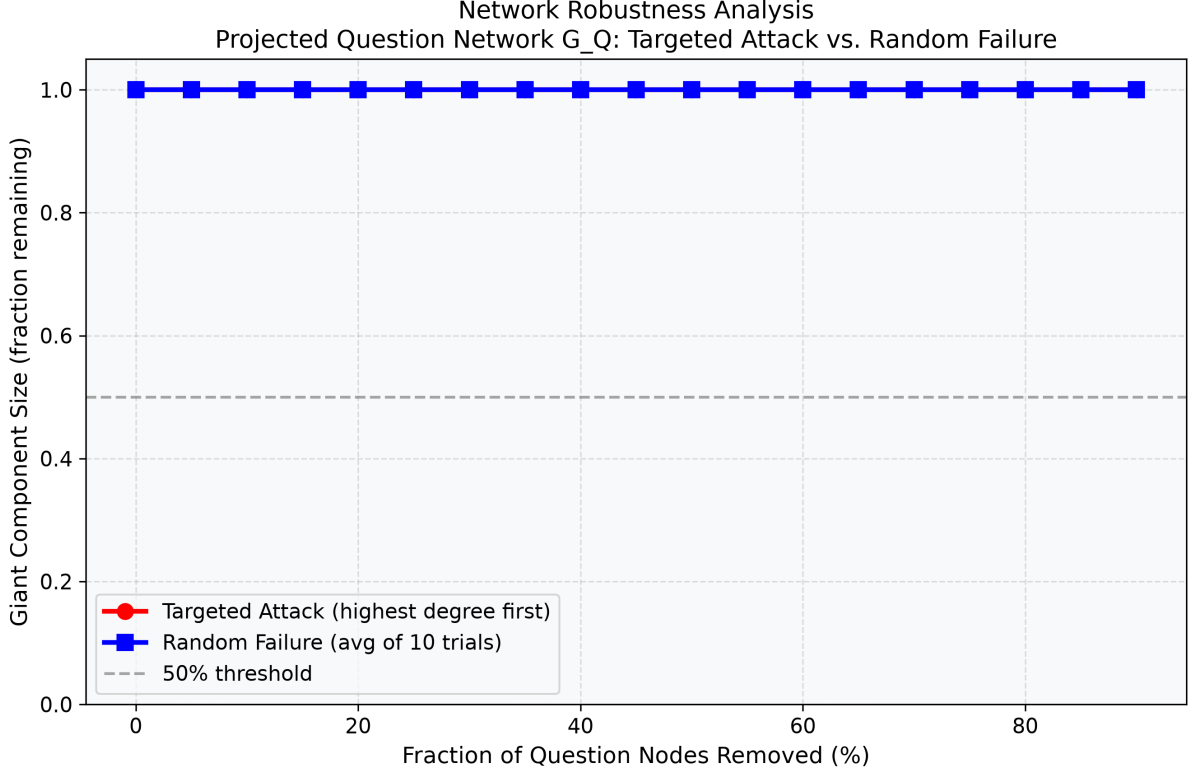


Figure 15: Robustness of the unweighted projection under targeted attack (highest degree first) and random failure. In  $K_{60}$  the two are mathematically indistinguishable and the giant component holds at 1.0 until the graph is nearly exhausted, because removing a node from a complete graph leaves a complete graph.

## 6 Results and Discussion

**1. The class agrees far more than it disagrees, and that shapes every network.** Four in five answers are on the agree side. In Network A this makes raw agreement worthless as a similarity measure — it links everyone to everyone — and forces the chance correction. In Network B the same fact appears as a bipartite density of 0.44, and in Network C as a complete projection. All three networks are dense for the same underlying reason.

**2. The three views agree on which questions matter, from opposite directions.** Network A finds E15, V13, S05, E11 and V09 to be the *most consensual* questions, the ones where matching answers carry almost no information. Networks B and C find exactly the same questions to be the strongest ideological anchors. There is no contradiction: these are the statements on which a large majority answers Strongly Agree, which makes them simultaneously the biggest gravity wells for intense opinion and the least informative for telling students apart. Continuous learning (E15) draws extreme views from 71.9% of the active cohort while adding almost nothing to any pair’s agreement score.

**3. Nothing fragments.** Network A’s largest component erodes one student at a time under thresholding, with no percolation-style collapse; Network B has a single connected component; Network C is complete, with  $\kappa = 59 \gg 2$  and no distinction between random failure and targeted attack. The cohort has no echo chambers, no ideological silos and no structural choke points. Whatever divides this class, it is not topic segregation.

**4. Structure is small-world, broad, but never scale-free.** Network A is clustered well above random ( $6\times$  in the sparse version) at random-like distances,  $\sigma = 1.46$  and  $5.66$ . Network

B’s question degrees are overdispersed (dispersion index 3.75) but bounded, with  $\hat{\gamma} = 0.750$  nowhere near the scale-free band. Both networks are broader than an ER graph and far from a power law — unsurprising at  $N = 91$  and  $N = 60$ , and worth stating as a limit rather than a discovery.

**5. Centrality in Network A largely measures agreeableness.** A student’s total agreement tracks their average answer at  $\rho = 0.89$ , and the emphatic answerers hold the heaviest edges. Degree and closeness instead reward students who mix Agree and Strongly Agree and sit between the emphatic and the cautious. Both readings are legitimate, but neither supports a claim about *which* opinions a central student holds.

**6. Agreement is topic-specific even though structure is not.** Society and Environment rank students most alike (pair scores 0.46, strengths 0.56); Technology and Education least (0.16). A student close to the centre of the environmental network may sit on the rim of the technology one. This is the clearest disagreement-shaped signal in an otherwise consensual class.

**Limitations.** The cohort is small (91 usable responses), so degree-distribution claims are bounded by  $N$ ; three respondents in Networks B and C answered only 15 questions, which understates their degree and inflates their betweenness in Network A; and a Likert survey cannot separate genuine conviction from a habit of reaching for the extreme end of a scale, so “emphatic” should be read as a response pattern, not a personality.

## 7 Individual Contribution

| Member      | Contribution                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mayank Kar  | Network A (student–student). Shared cleaning and encoding module <code>clean.py</code> used by the whole team; the simple and chance-adjusted agreement scores; the sparsification experiments (global threshold sweep and the top-2 construction); the structural measurements against ER graphs, including clustering, path lengths, efficiency, small-world $\sigma$ and assortativity; all five centrality measures and their rank comparison; the per-topic networks; <code>student_network.ipynb</code> and the eight figures in <code>student_plots/</code> . |
| Tanush Garg | Network B (student–question bipartite). The extremes filter and long-format edge list; construction and validation of the bipartite graph; bipartite topology (density, components, isolated respondents); question-node degree distribution with linear and logarithmic binning, the power-law fit and the dispersion analysis; the bipartite layout figure; <code>main.ipynb</code> and the LaTeX sub-report for the extremes network.                                                                                                                             |
| Amish Goyal | Network C (question–question projection). Projection of the bipartite graph onto the question set and the weighted projection $G_Q^W$ ; strength, weighted betweenness and weighted eigenvector centrality; the role-based classification of anchors and bridges; the Molloy–Reed evaluation and the robustness simulation (targeted attack versus random failure); the projection, centrality and robustness figures; repository setup and the interactive D3 visualisers.                                                                                          |